

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester IV Examination 2025-26 (NEP)

CHEMISTRY**Paper Code: CHEMJ41 : Inorganic Chemistry-II**

Ref. No.: Conf/Even/N/2025-26/05

Time: 2½ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1.

- (a) CsI is much less soluble in water than CsF and LiF is much less soluble than LiI. Why? [2]
- (b) Why is $[\text{Ni}(\text{en})_3]^{2+}$ more stable than $[\text{Ni}(\text{NH}_3)_6]^{2+}$ though both are octahedral complexes? [2]
- (c) What is the difference between solvation and solvolysis? Explain with one example of each type. [2]
- (d) What is E-factor in green chemical processes? [2]
- (e) What is lanthanide contraction and what are its important consequences? [2]
- (f) Write the names and electronic configurations of two naturally occurring actinide elements. [2]
- (g) What is Hammett acidity function? [2]

Question 2.

- (a) Draw the various stereoisomers and specify the enantiomeric pairs in (i) Ma_4b_2 and (ii) $\text{M}(\text{AA})_3$ type of octahedral complexes. [6]
- (b) State the properties that determine the utility of a solvent. [4]
- (c) What are the differences in characteristic properties of blue colored solution and bronze colored phase obtained by addition of alkali metals in liquid ammonia? [4]

Question 3.

- (a) Briefly write about the various principles of green chemistry. [6]
- (b) Arrange the following acids in decreasing order of acid strength in aqueous solution. Explain the reason behind it: H_2SO_3 , H_2SO_4 , HMnO_4 , H_3AsO_4 . [2]
- (c) Give two examples each of polar aprotic and polar protic solvents and write autoionization reaction for them. [3]
- (d) Predict the product when $(\text{CH}_3)_2\text{N-PF}_2$ is reacted with (i) B_2H_6 and (ii) BF_3 . Explain the reactivity difference in the two cases. [3]

Question 4.

- (a) Using valence bond theory, predict the structure and spin only magnetic moment of (i) potassium tetracyanonickelate(II) and (ii) hexaamminecobalt(III) chloride. [4]
- (b) Write the electronic configuration and number of unpaired electrons in the ground state of following ions: Tb^{4+} , Gd^{3+} , Ce^{4+} , Lu^{3+} . [4]
- (c) Explain the reason behind following observations: [4]
- (i) Silver chloride is colorless while silver iodide is yellow in color.
- (ii) $[\text{CoCl}_4]^{2-}$ is more intensely colored than $[\text{MnBr}_4]^{2-}$.
- (d) Complete the following reactions and identify acid and base in each case: [2]
- (i) $\text{NH}_2\text{SO}_2\text{OH} + \text{H}_2\text{O} \rightarrow ?$
- (ii) $\text{CaO} + \text{SiO}_2 \rightarrow ?$

Question 5.

- (a) Identify and draw the various structural isomers of the following complexes: [6]
- (i) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$
- (ii) $[\text{CoBr}(\text{NH}_3)_5]\text{SO}_4$
- (iii) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
- (b) What are the limitations of effective atomic number rule? [2]
- (c) What are the various methods of separating the lanthanides? [6]

Question 6.

- (a) What are intramolecular and intermolecular frustrated Lewis acid-base pairs? Explain with an example of each type. [4]
- (b) Explain the following with the help of suitable examples: [8]
- (i) Advantage of solvent free reaction in green chemical synthesis.
- (ii) Difference between ordinary acids and superacids.
- (c) The most common source of mercury is cinnabar (HgS), whereas Zn and Cd in the same group occur more commonly as carbonate, silicate and oxide. Why? [2]

Question 7.

- (a) Briefly explain the various methods of uranium enrichment. [6]
- (b) What are ionic liquids? Briefly classify them. [4]
- (c) Write the IUPAC names of the following: [4]
- (i) $\text{K}_3[\text{Fe}(\text{CN})_6]$
- (ii) $\text{K}[\text{Cr}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]$
- (d) Draw the structure of: $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ and $\text{Fe}_2(\text{CO})_9$.
